

Digital Twins for Civil Infrastructure

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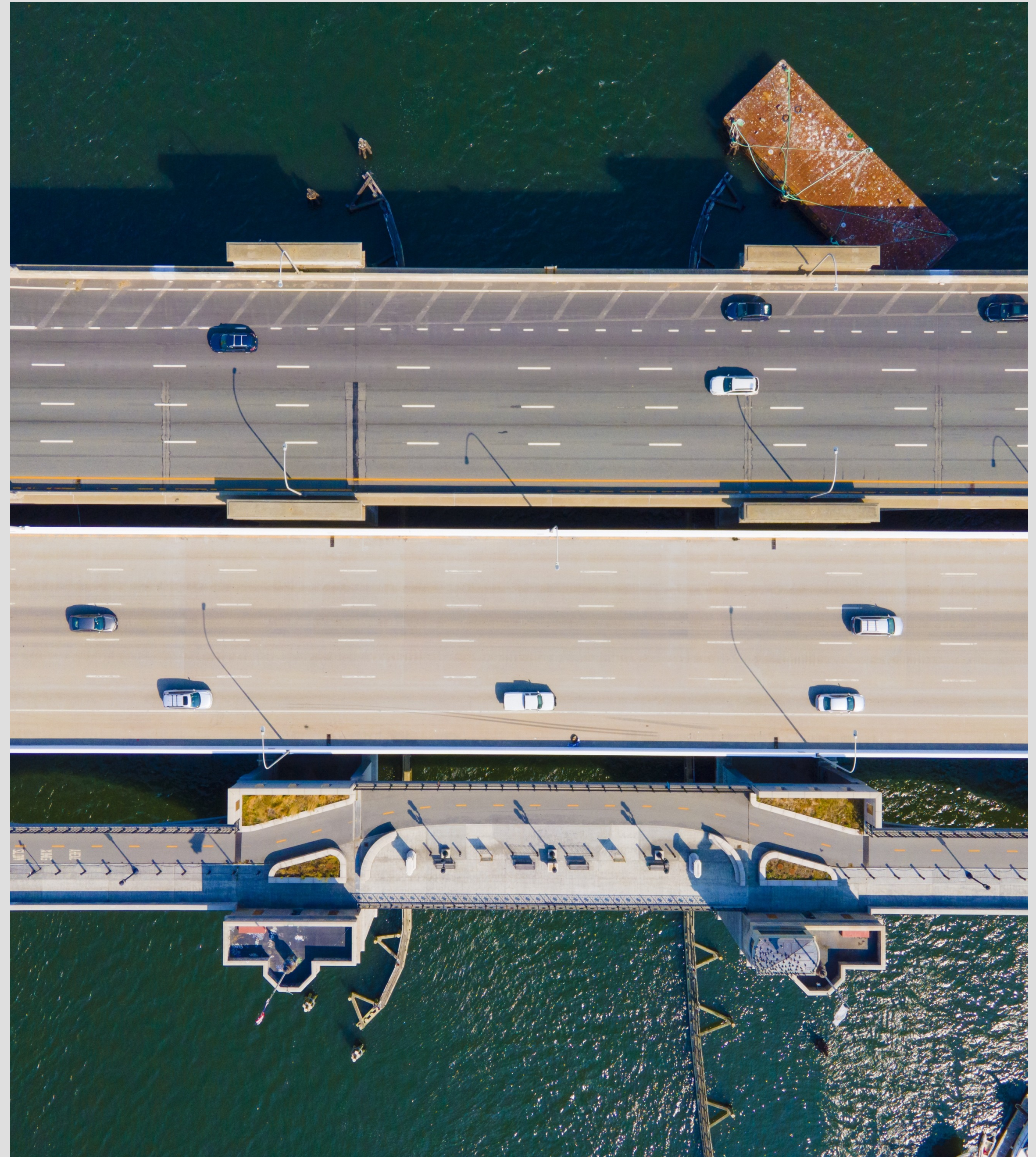
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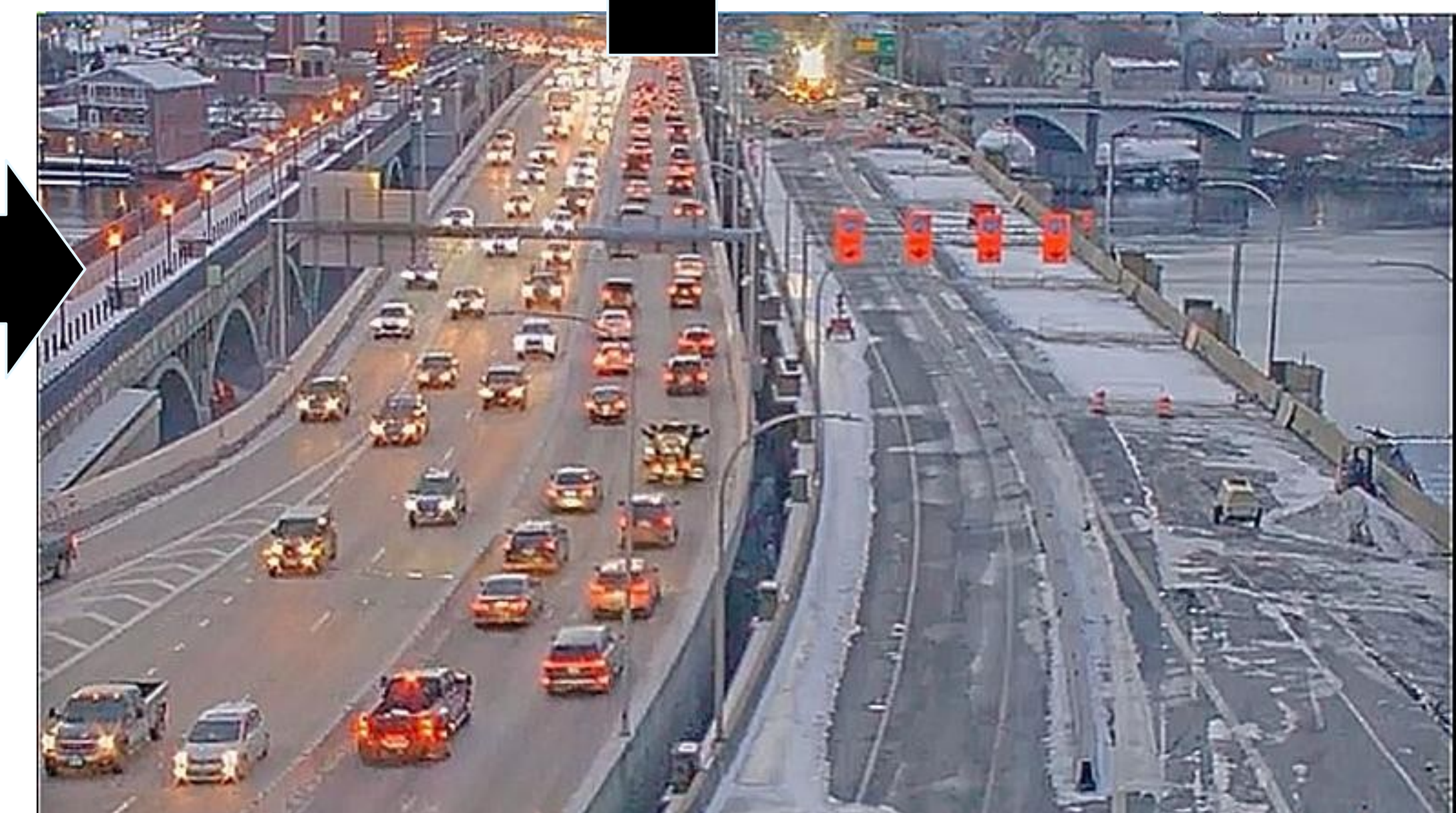
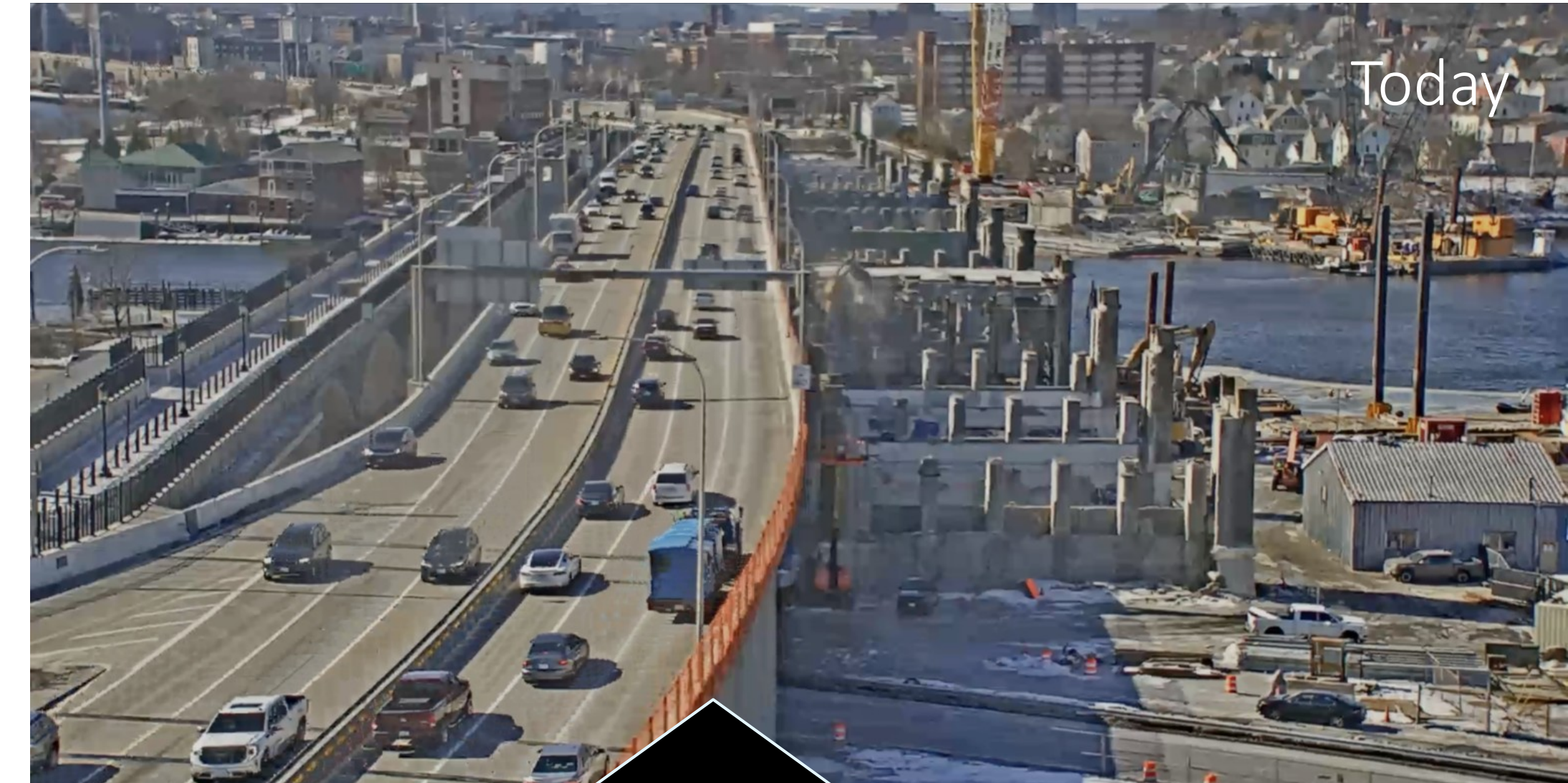
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Challenges with Bridge Inspection

Issues with structures can be hard to catch with visual inspection.

- Metal rebar embedded in concrete can corrode without presenting obvious signs externally
- Even external elements can be obscured by the way structures are designed
- Natural materials can build up, further obscuring key elements
- Concerns about drone chip sets may further limit visual inspection



Challenges with Managing Inventory

- Nearly the highest number of bridges per capita in the country
- Rhode Island has more large structures over water
- Most expensive assets to maintain
- Hard to monitor and know when the best time to intervene to extend life
- Aging inventory: large number of assets >30 years
- Changing conditions – temperature, storms, traffic and vehicle mix



State	Bridges	Population	Bridges per Capita
Kansas	24,948	2,970,000	0.00840
Rhode Island	782	1,110,000	0.00070
Delaware	874	1,060,000	0.00082
Wyoming	3,138	588,750	0.00533
Alaska	1,685	738,737	0.00228
Florida	13,070	23,500,000	0.00056
New York	2,000	20,000,000	0.00010
Massachusetts	5,245	7,000,000	0.00075
Virginia	11,900	8,800,000	0.0014
Connecticut	4,200	3,688,496	0.0011

Key Questions

- **How can RIDOT monitor the bridges for structural health as they carry more traffic?**
- **How can RIDOT tell what the sensor readings mean, and how can we tell what is normal vs. an important change?**
- **If an important reading is identified, what should be the course of action?**
- **Can the system be extended to monitor more assets across the state?**

2. Alternative Routes

Increased load as traffic seeks different paths

Recommendations

- Deploy sensors into asset management system for monitoring
- Model and monitor traffic variances from baseline

1. Bridge Monitoring



Traffic doubled up on remaining structure will accelerate impact, posing a higher risk to a structure that now has higher visibility

Recommendations

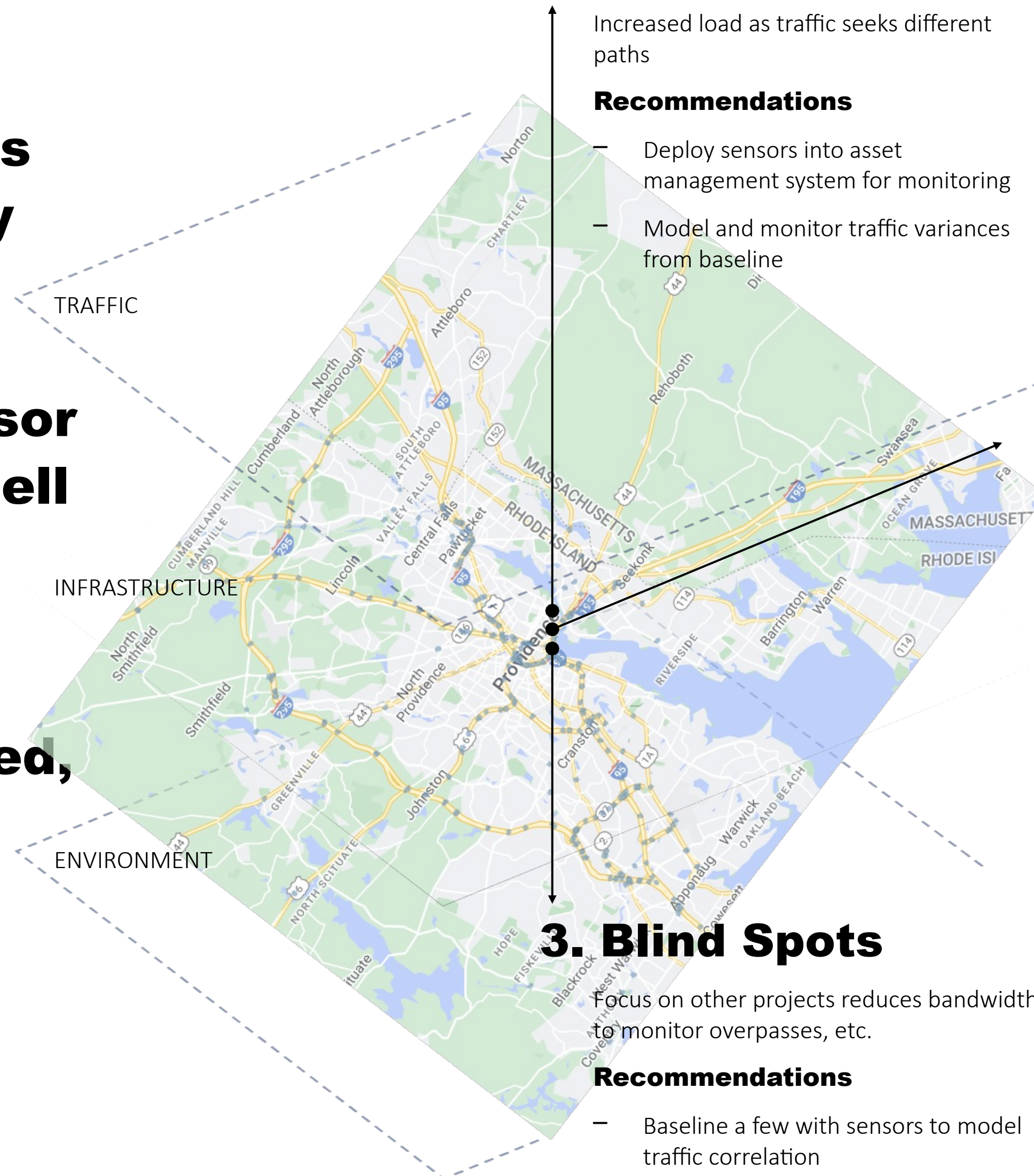
- Fiber sensors along structure to establish baseline performance and enable ongoing monitoring
- Deploy drone monitoring on more frequent basis than manual checks to look for visual changes that may indicate a problem

3. Blind Spots

Focus on other projects reduces bandwidth to monitor overpasses, etc.

Recommendations

- Baseline a few with sensors to model traffic correlation
- Use traffic to monitor risks



Case Study

Autostrade per l'Italia

This client is a joint-stock company that operates a number of roads and bridges across Italy. In this role, the company is the largest operator of tolled assets in the country, including over 4,000 bridges.



Infrastructure Asset Management

Problem

Create a centralized system to keep an eye on all assets operated, while providing a central way to manage maintenance.

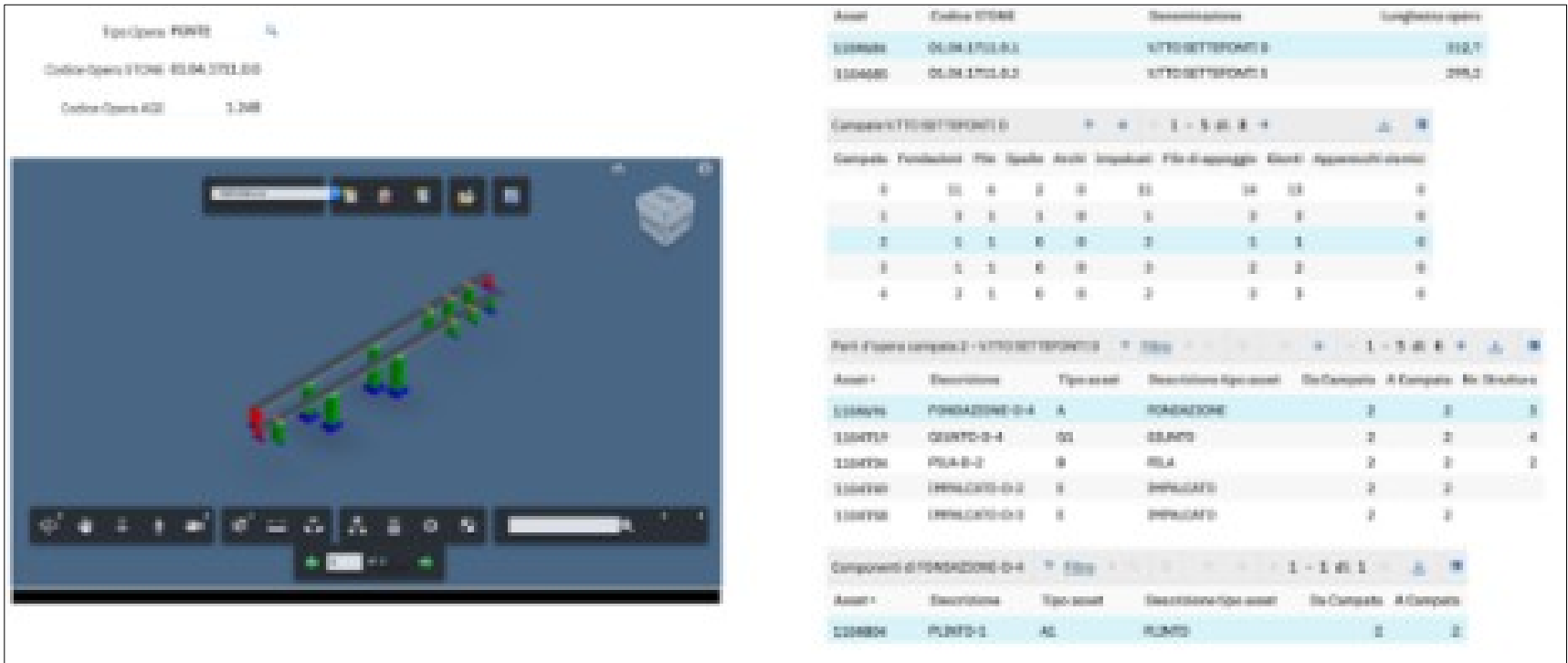
Solution

Working with an IBM partner, the client implemented IBM Maximo for Civil Infrastructure, configured to support the company’s new asset management program, connected to data from sensors and equipment.

Impact

- Improved visibility with a centralized digital twin monitoring all assets]
- Better insights about each structure based on data feeds
- Optimized work management
- Improved public confidence

Drones create digital twin



Sensors add data connected to the model



AI generates insights

WIND | NRDP

WIND | RP10

WIND | VROZ

NRDP: 9.13 m/s (5 BFT)

gust: 11.38 m/s

197.1°

Last update: 22:45:25 EDT

PHW and PLW | HOEK

PHW and PLW | BOTNM

	HOEK	cm (ALAT)
PHW	We 23:00	255
PLW	Th 07:00	48
PHW	Th 11:30	231
PLW	Th 16:20	31
PHW	Th 23:40	279
PLW	Fr 07:50	88

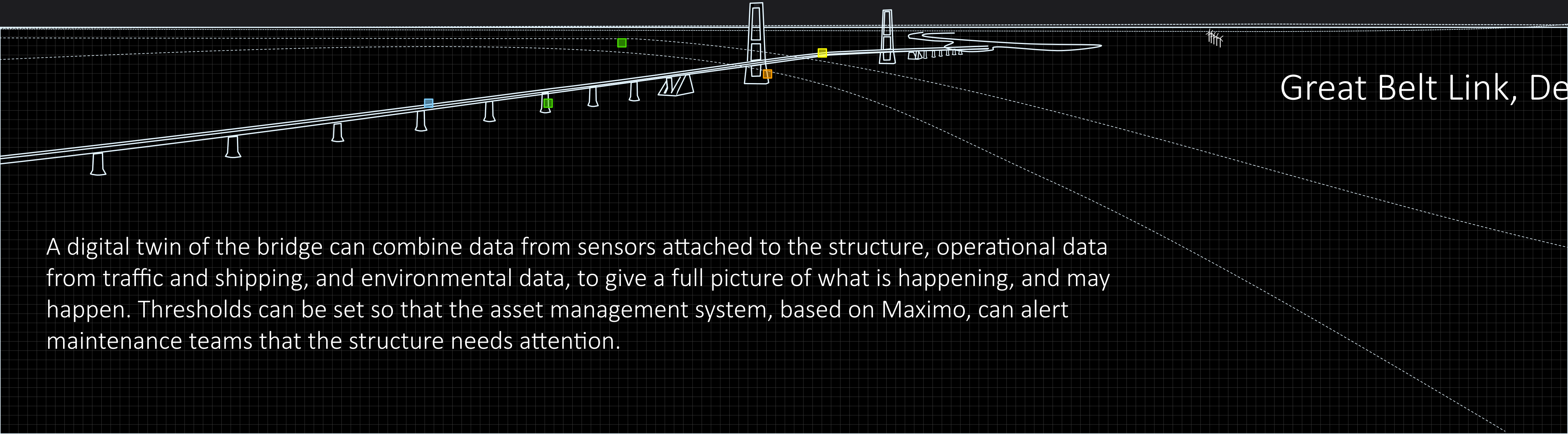
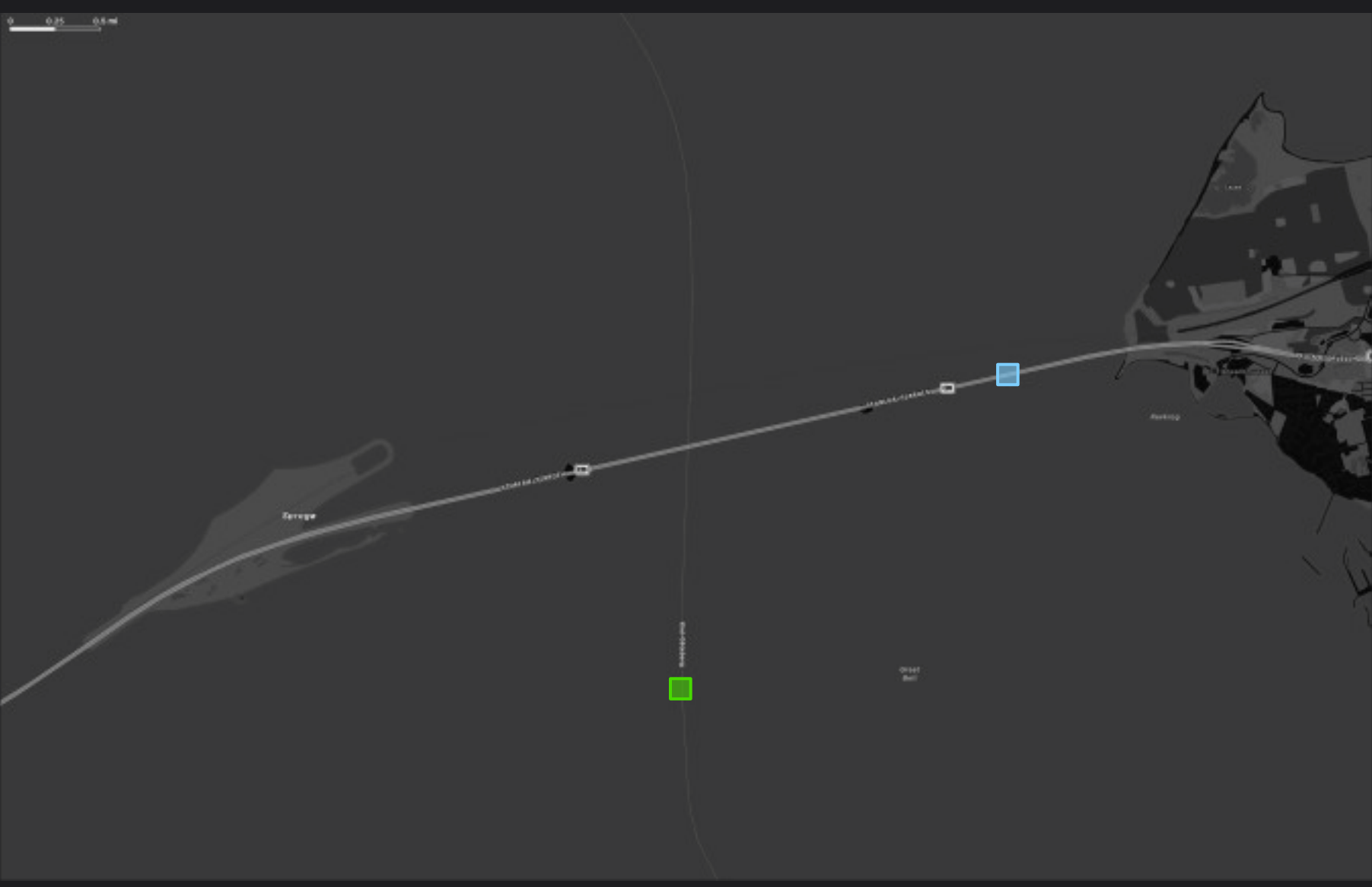
PSLW | HOEK

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PSLW | BOER

	HOEKNWW
PSLWDA	We 13:14
PSLWDA	We 21:31
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PSLWDA	Th 10:02

Last update: 22:45:26 EDT



Great Belt Link, De

A digital twin of the bridge can combine data from sensors attached to the structure, operational data from traffic and shipping, and environmental data, to give a full picture of what is happening, and may happen. Thresholds can be set so that the asset management system, based on Maximo, can alert maintenance teams that the structure needs attention.

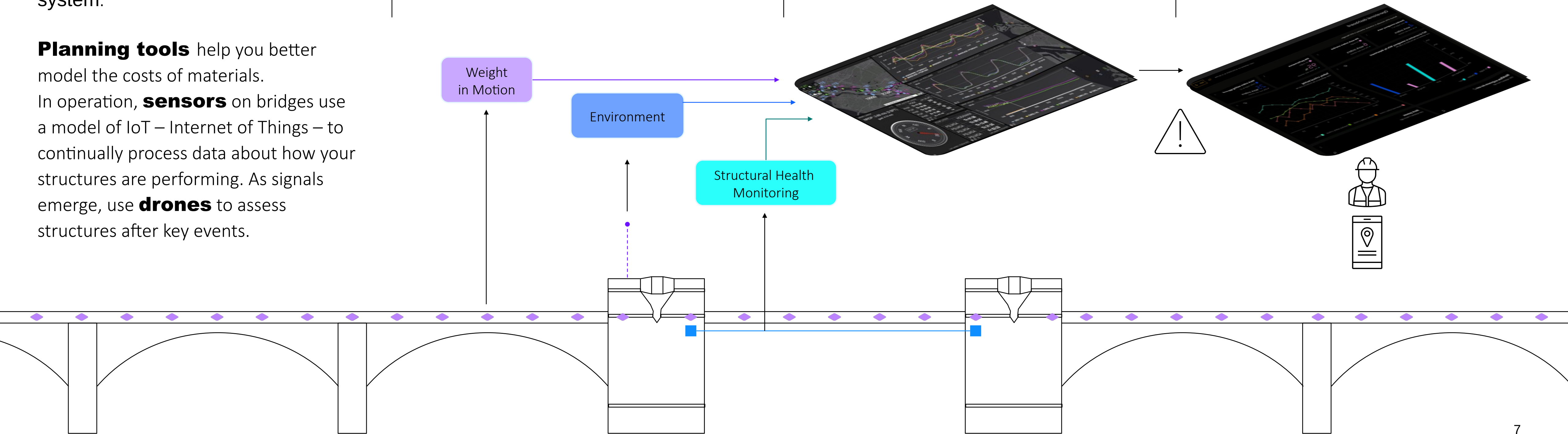
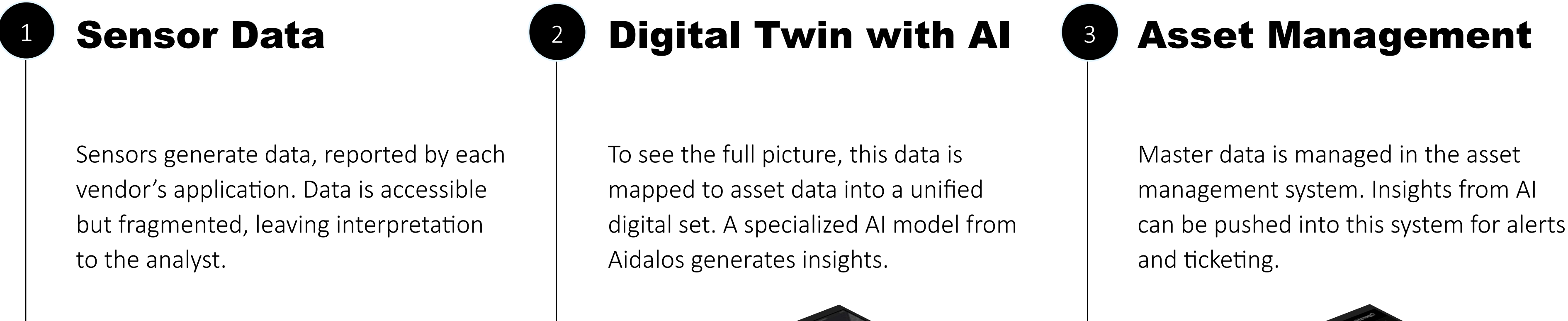
Test Bridge

Digital Twin

Enable shift from reactive to proactive

Aging infrastructure, combined with changing patterns and more frequent weather events, drive the need for an **infrastructure management system**.

Planning tools help you better model the costs of materials. In operation, **sensors** on bridges use a model of IoT – Internet of Things – to continually process data about how your structures are performing. As signals emerge, use **drones** to assess structures after key events.



Manage

gtmdemo.manage.wwdemo1itz.gtm-pat.com/maximo/oslc/graphite/manage-shell/index.html?event=loadapp&value=plusxasset&uisessionid=1927&_tt=qpffcuajgm3rkb3jf5gl26b35k#/m...

IBM Maximo Application Suite

Manage

Take a tour

Assets (Civil)

List View

Find navigation item

Available Queries

Common Actions

New Asset

Save Asset

Clear Changes

Change Status

Move/Modify Assets

Swap Assets

Associate Users and Cust...

Create Report

Unlink GIS Feature

Application Import

Application Export

Import Digital Data

Delete linked GIS Feature

More Actions

Define Maintenance Schedule

Manage Maintenance Schedule

Define Operational Schedule

Manage Operational Schedule

Asset

Spare Parts

Safety

Meters

Specifications

Features

Relationships

Work

Work Zones

3D View

Service Address

Map

Asset

PILLAR103

EB Pillar 103

Site

BRIDGE

View Work Orders and Tickets that are open for this asset. Additional details are available in Work Details.

View Work Details

Work Orders

Tickets

Defects

Defects

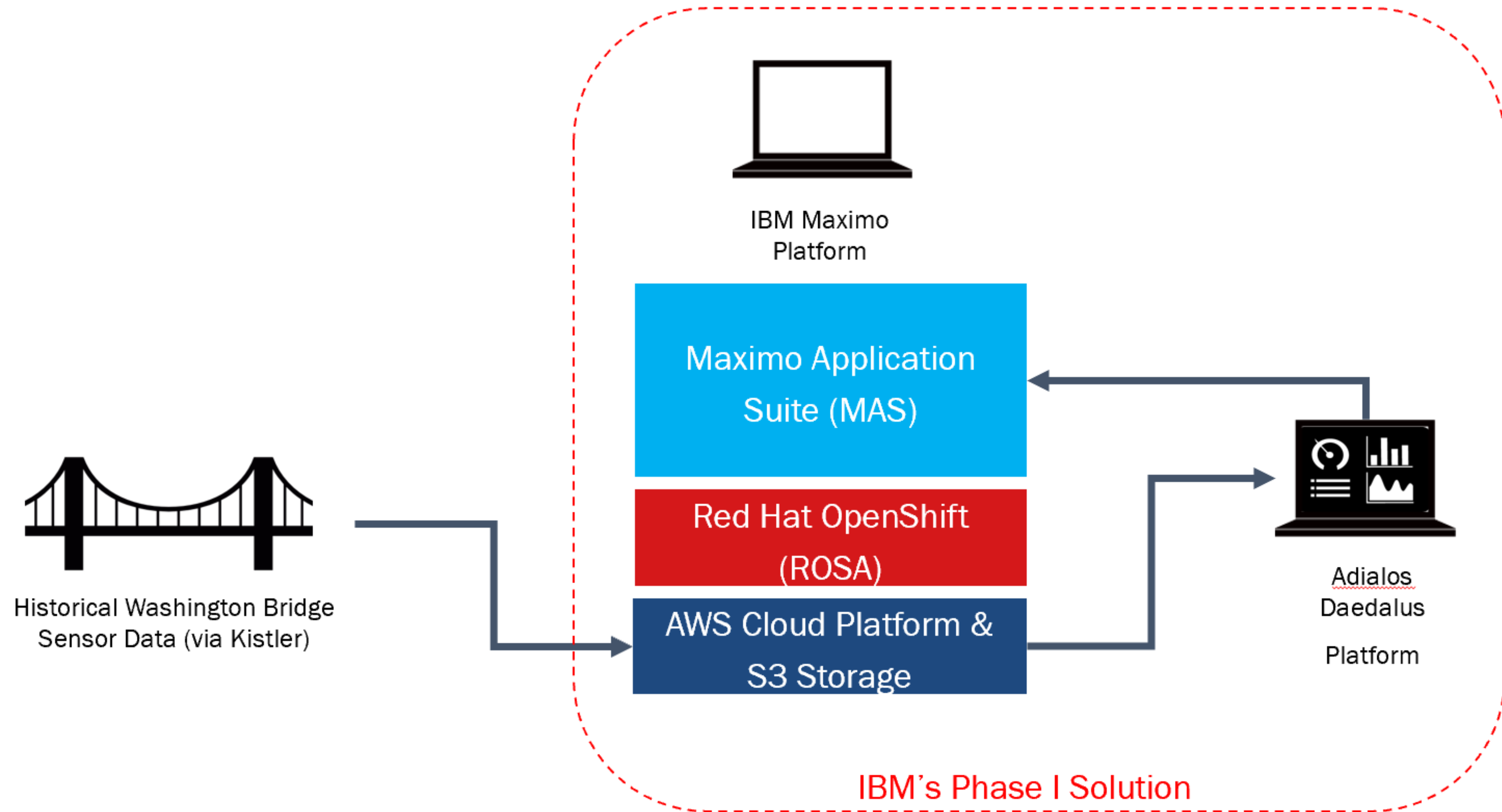
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Refresh Defects

Defect		Summary	Type	Cause	Score	Status	Has Markup?	
1179	>	Algae on West face at water line	ALGEA		3	QUEUED	☐	
1180	>	Cracks in pillar	CRACK		2	QUEUED	☐	
1185	>	Spalling leading to crack in structure	SPALLING		2	QUEUED	☒	
1198	>	Crack in pillar	Crack	Weather	2	WOR	☐	
1199	>	Cracks in pillar	Crack	Weather	2	PENDING	☐	
1200	>	Cracks in North Face, Middle	Cracks	Weather	2	PENDING	☐	
1212	>	Crack in north face, middle top	Crack	Weather	2	PENDING	☐	
1213	>	Cracks in Eas Face lower corner	Cracks	Weather	5	PENDING	☒	
1242	>	Structural damage on wall	Vertical Crack	Shift in ground due to weather event (flooding)	4	NEW	☐	
1245	>	Large crack caused from base erosion			3	NEW	☐	

Maintenance workers can receive a prioritized list of issues to review, each with a work ticket that contains the history for them to review.

Architecture

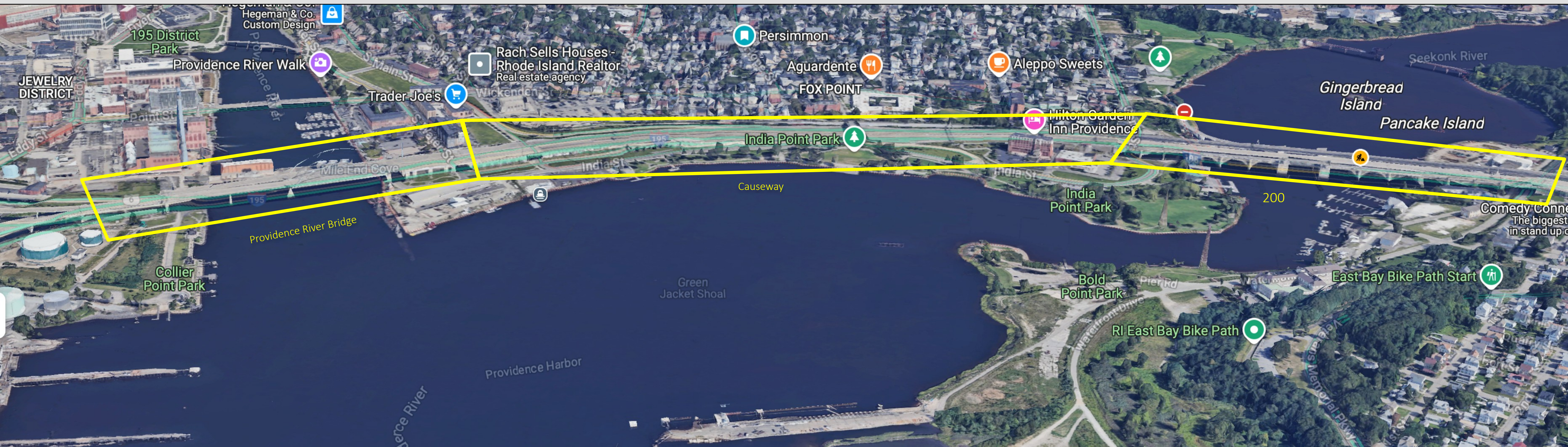


Corridor Monitoring

Integration of assets across corridors

What happens to one asset in the corridor can affect the others. Traffic patterns change, increasing or decreasing loads over time, by the hour, until new configurations and structures are in place.

Sensor readings from one structure can predict the load the other structures will handle and provide correlations that can be used to compare how structures may degrade over time, to improve forecasting and planning.



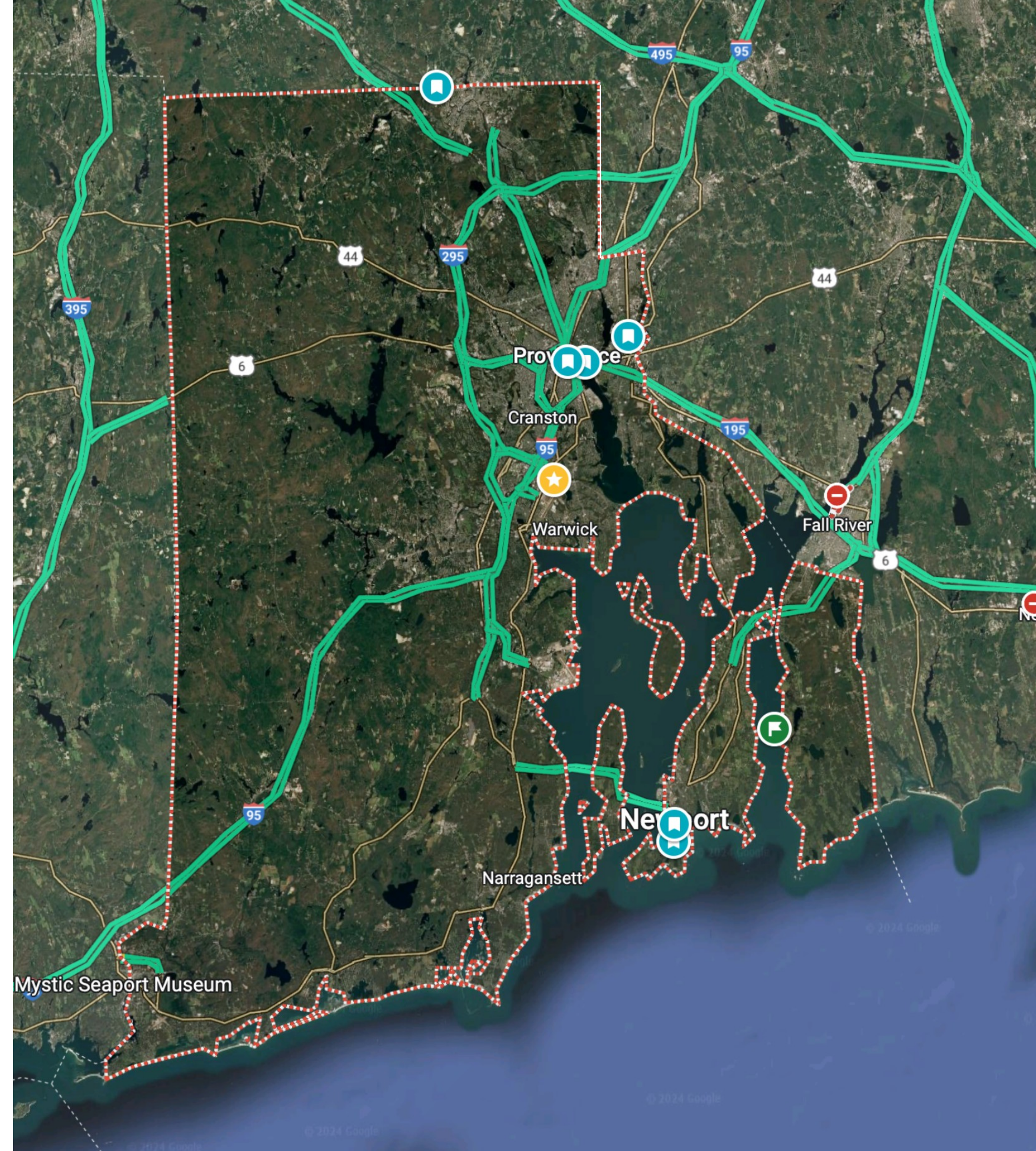
Wide Area Monitoring

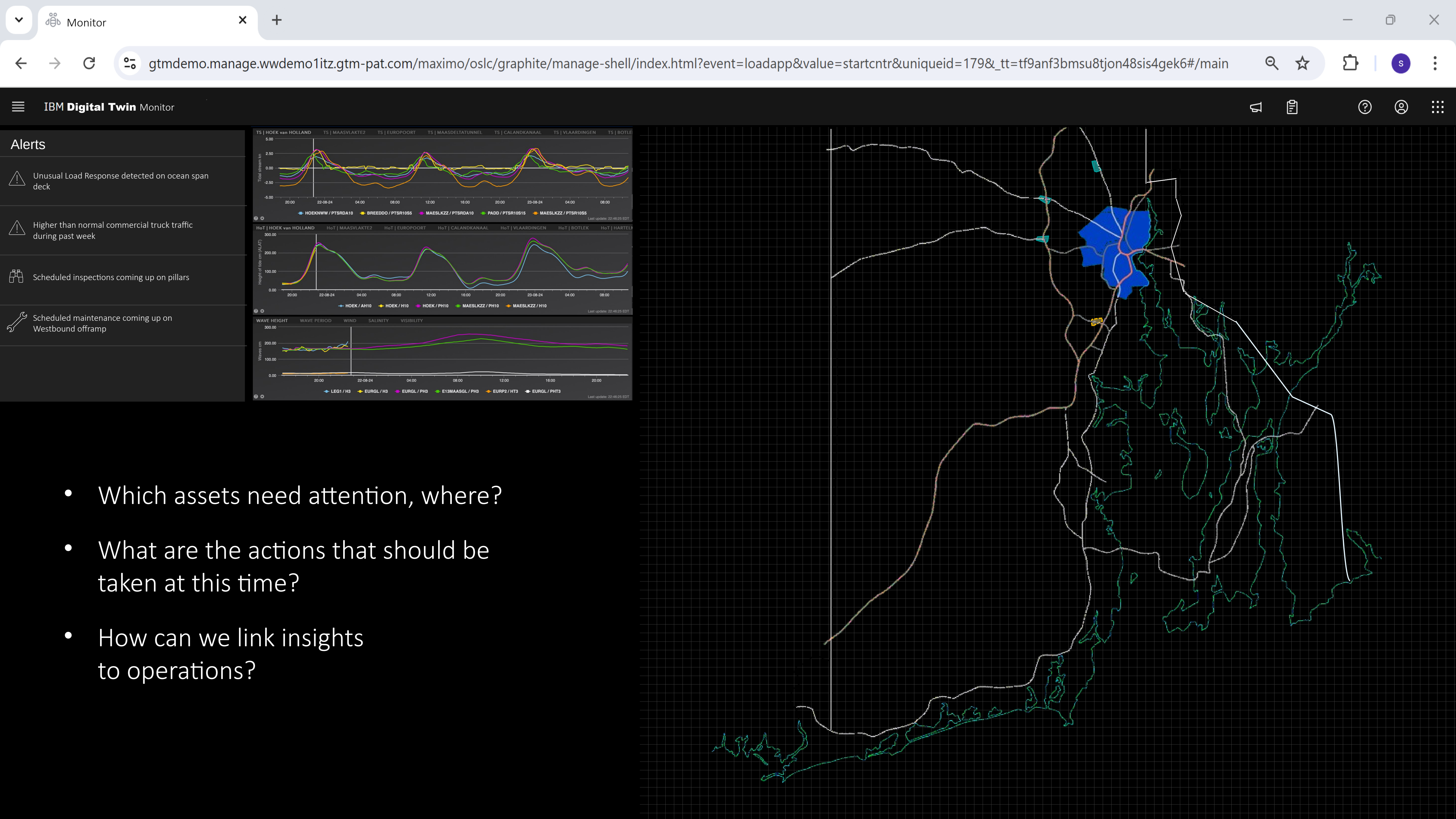
DOTs have a significant amount of assets they are responsible for, over a very wide area.

It's hard to know what's happening with these assets, short of going around and checking them manually.

Even when checking manually, it can be hard to find out how the structures are responding to the loads they are handling.

With a central system, data from these structures can be monitored, so alerts can be shown to where attention is needed.





Thank you

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